

PID-RS RESEARCH PROGRAM

PID discovery, verification, and durability blueprint

PrimeGaps lessons, semantic-transfer firewalls, exact SxPID3 assurance, autoresearch,
and durable promotion

A source-pinned, 20-lens review of proof engineering, exact categorical SxPID3,
autoresearch promotion, formal-method limits, and durable repository practice.

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1 Lessons from PrimeGapsLib, transferred without semantic conflation

2 Status and scope

This document is a source-pinned research and engineering review, not a theorem and not an acceptance receipt. It studies Axiom Math's PrimeGapsLib at exact Git commit 1faa7b14e82ddebcb2772dfb9153922f01b106477, the interactive paper and its 132-page blueprint, then asks which proof-engineering patterns can improve pid-rs discovery and verification.

The answer is deliberately asymmetric:

- several proof-engineering structures transfer usefully;
- the prime-gap mathematics does not transfer to partial information decomposition;
- an exact finite certificate does not validate a continuous estimator, a population model, a statistical procedure, or a scientific interpretation; and
- formal verification is one instrument in an argument, never a replacement for semantic review, mathematical judgment, falsification, numerical analysis, or independent custody.

The principal positive design is scoped to the proposed three-source empirical categorical shared-exclusions packet SX-CERTIFIED-AVERAGED-PID3-001. It introduces a typed semantic mass DAG, a separate reduced-rational product plan, forward and backward sparse-completeness checks, and a content-addressed sharded replay. This is a project-defined assurance architecture for an existing paper-defined functional. It is not a new PID measure or a currently accepted certificate.

The process is part of the result. This document therefore records the source inventory, formulas, theorem chain, 20-lens council, dissent and limitations, autoresearch boundary, implementation roadmap, evidence hierarchy, and repository durability state machine. Exact path inventories, hashes, run rosters, and mutation records belong in machine-readable JSON or TSV companions rather than being copied repeatedly into human-facing migration PDFs.

2.1 Publication contract

This Markdown file is the canonical human narrative. The paired PDF is a derived publication, not a second editable source. Run scripts/build-pid-discovery-verification-blueprint.sh to regenerate it. The builder stages only this Markdown, its formatting filter/header, and the two handcrafted SVGs; fixes locale, time zone, creation epoch, document fonts, and temporary state; performs two isolated three-pass LuaLaTeX builds; rejects compiler/layout warnings, non-A4 output, missing semantic text anchors, and unembedded or non-Unicode-mapped fonts; and requires the two PDF byte strings to match.

That equality is a **same-toolchain observation**, not a universal cross-platform theorem. A new Pandoc, LuaLaTeX, SVG renderer, font stack, or operating-system backend must rerun the gate and may legitimately produce different bytes. Cross-toolchain acceptance requires a separately specified structural comparison and visual inspection. The transfer ledger remains the machine-readable authority for exact source hashes, anchors, semantic cards, and the 20-lens registry; neither the PDF nor its polish upgrades the scientific status of the proposed SxPID3 packet.

3 Executive decision

3.1 Adopt

1. **Same-revision theory, certificate, and composite lanes.** Keep abstract mathematics, finite exact evidence, and the theorem that combines them separate, but require them to refer to the same immutable source revision.
2. **Untrusted discovery followed by a smaller checker.** Numerical search, autoresearch, and code generators may propose candidates; acceptance reconstructs the mathematical object from frozen input and checks it with a smaller, separately reviewed route.
3. **Explicit finite-to-semantic bridges.** A checked integer or rational inequality becomes useful only through named lemmas that connect stored data, mathematical definitions, and the final conditional claim.
4. **Assumptions in theorem and report types.** External premises, support conditions, units, estimand identity, and conditional status remain visible instead of being shortened away in summaries.
5. **Human-readable challenge plus machine comparator.** Freeze a small theorem roster, import and axiom policy, and an optional second-kernel replay independently of the candidate implementation.
6. **Typed dependency graph and completeness checks.** Sparse or packed evidence needs both directions: every required object is represented, and every represented object is semantically valid.
7. **Content-addressed shards and promotion receipts.** A large finite replay is a partitioned certificate family, not an opaque long job or an expiring CI artifact.

3.2 Adopt conditionally

1. Packing, custom kernels, and generated tables only after a simple streaming implementation is correct and profiling shows a real bottleneck.
2. A comparator only after its binary, version, theorem roster, allowed axioms, inputs, and full output are pinned and replayed in CI or under independently controlled custody.
3. A factor DAG inside SxPID3 revision 1 only if it leaves the frozen certificate semantics and resource admission policy unchanged. If the graph becomes an acceptance-bearing public object or changes an admission limit, the claim needs a new adjudicated revision.

3.3 Reject or defer

1. Direct transfer of GPY/Maynard sieve formulas, the number 246, the 50-tuple, factorial moments, or Bombieri–Vinogradov reasoning to PID.
2. Any implication from a finite exact categorical certificate to KSG consistency, continuous support, bias, variance, confidence coverage, population PID, causality, or scientific novelty.
3. Treating a green cached build, model consensus, self-assessed formalization, source hash, or polished blueprint as sufficient independent review.
4. Treating mutable blob/main source links or a status-colored dependency graph as a durable proof-to-paper correspondence receipt.
5. Kernel evaluation of millions of SxPID3 verdicts before a general proof and a bounded, independently replayable executable checker exist.

4 Epistemic contract

Every proposed transfer is represented by seven fields:

1. source meaning;
2. PID analogue;
3. assumptions preserved;
4. assumptions changed or introduced;
5. prohibited inference;
6. evidence required before adoption; and
7. disposition.

The complete machine-readable form is `audit/evidence/primegaps-to-pid-transfer-ledger-v1.json`. Shared words such as *information*, *lattice*, *factor*, *kernel*, *moment*, *certificate*, *positive*, or *independent* do not create a mapping theorem. Context determines meaning.

The council is advisory. Reviewers are asked to produce falsifiers and corrections, not votes. The integrator must reconstruct decisive claims from primary sources and owns the final scope. Two agents in one environment remain correlated evidence even when they use different roles or prompts.

4.1 What the tools can and cannot establish

Tool or method	Useful conclusion	Boundary that remains
Lean kernel	A term has the declared type under the recorded imports and axioms.	It does not prove that the formal statement faithfully represents the intended scientific claim.
Exact integer/rational checker	A finite algebraic predicate holds for reconstructed exact inputs.	It does not establish sampling validity, a population parameter, continuous support, or external provenance.
SMT solver	The bounded encoding is unsatisfiable or has a model under the encoded theory.	A missing premise, wrong encoding, or unsupported unbounded generalization remains possible.
Directed interval arithmetic	The declared exact real lies inside a rigorously constructed interval under its runtime assumptions.	It is not a confidence interval and does not establish model or estimator validity.
Mutation suite	Named faults are detected by the acceptance predicate.	It does not prove that all relevant faults have been enumerated.
CI	One bound revision passed one recorded environment and command family.	It does not prove universal portability, fresh reproducibility, or scientific correctness.
SHA-256	Two byte strings are equal except with the hash's cryptographic collision risk.	It does not establish truth, authorship, authenticity, semantic identity, or durable availability.
Multi-agent council	Diverse objections and candidate routes can be generated efficiently.	Shared training, prompts, sources, tools, custody, or objectives can create correlated blind spots.

5 Exact source inventory

The review uses the following identities. Reviewed means the bytes or Git object were inspected; it does not mean every statement was independently reproved.

Object	Exact identity	Review use
PrimeGapsLib tree	commit 1faa7b14e82ddeb2772dfb9153922f01b106477; tree 611f4b1caccb7850996cf425aefb1a6c42ed629f	Lean source, dependencies, comparator, history, and CI architecture
Interactive paper HTML	SHA-256 8ab349b59a897780ee7b023ec9b350442fc77b36aec84873019bfff2612a97c90; 10,793,548 bytes	statement/proof dependency graph, status, and source-link audit
Blueprint PDF	SHA-256 c93efd077ff4236926379d9879cc82e124a12f4eccc1c43a685054b6ba866117; 1,153,093 bytes; 132 A4 pages	human proof narrative and visual inspection
Blueprint layout text	SHA-256 1ea1f0fb79561a91edba050bc767325882aee83c51dd1ba169e2e0a5e25bd8cf; 495,392 bytes	searchable formula and statement review
Certificate source JSON	SHA-256 99808e1d95204c5bb0cde845641cfdbb41ea950fa7fa9858df8bd093111787a; 55,303 bytes; 1,295 rows	untrusted candidate data input
Reviewed CI job log	SHA-256 408c6b2193069268c2250227e326d0aabdd233244b1b6de804c9c98d4750c2b5; 33,294 bytes	observed warm-cache build behavior and skipped assurance routes

Primary locators are PrimeGapsLib, the interactive paper, and the blueprint PDF. Repository links in this review use the exact commit rather than mutable main.

The non-Git web/PDF/log rows above are **hash-only review observations**: their bytes are not shipped in this repository. Their digests neither provide a recovery route nor authenticate the publisher, and a future reacquisition may differ. The companion ledger marks that custody limit explicitly; the commit-pinned Git source is the only source family in this report with a direct object-recovery locator.

6 What PrimeGapsLib proves

The headline result is conditional. The paper states the result using the familiar liminf formulation

$$\liminf_{n \rightarrow \infty} (p_{n+1} - p_n) \leq 246.$$

whereas the reviewed public Lean declaration keeps the Bombieri--Vinogradov premise explicit and uses a frequently-at-infinity formulation:

```
bombieriVinogradov_implies_frequently_prime_gap_le_246
(hBV : BombieriVinogradov) :
exists frequently m atTop,
```

```
nth Prime (m + 1) - nth Prime m <= 246
```

The two formulations belong to the same mathematical narrative, but this review does not silently substitute one theorem statement for the other. Neither may be summarized as an unconditional proof supplied by the certificate alone.

6.1 Mathematical skeleton

For an admissible tuple and nonnegative sieve weights, the analytic argument studies a difference of weighted sums of the form

$$S(N, \rho) = S_2(\lambda) - \rho S_1(\lambda).$$

The finite numerical route supplies a strict variational witness for

$$M_{k,\varepsilon} = \sup_F \frac{\sum_{m=1}^k J_{k,\varepsilon}^{(m)}(F)}{I_k(F)}.$$

At $k = 50$ and $\varepsilon = 1/25$, the repository represents a polynomial candidate as

$$F_c = \sum_i c_i b_{a_i, \alpha_i}$$

and checks, after clearing denominators, the exact inequality

$$4 \sum_{i,j} c_i c_j J_k^\varepsilon(i, j) < k \sum_{i,j} c_i c_j J_k^\varepsilon(i, j).$$

The exact Lean certificate conclusion used here is the strict threshold $M_{50,1/25} > 4$. The blueprint discusses a numerically constructed rational vector with an exact quotient above 4.0043, and a separate Lean file verifies that the parameter $Q = 4.004306$ makes a named positivity window nonempty. This review found no Lean theorem joining that stronger numeric margin to the stored certificate. It therefore does **not** describe the packed certificate as kernel-checking $M_{50,1/25} > 4.0043$.

Likewise, the blueprint's full degree-27 basis has 2,526 elements, while the reviewed JSON contains 1,295 nonzero sparse labels. The JSON row count is not the full basis size. The blueprint's degree-through-24 numerical observation is not presented here as a formal optimality theorem.

The numerical search that found the coefficient vector is not part of the trusted conclusion. The accepted route checks the exact integer/rational predicate, bridges the stored tables to the explicit quadratic forms, converts the explicit polynomial to an analytic L2 witness, derives $M_{\{50,1/25\}} > 4$, combines it with the analytic distribution premise, and uses an admissible 50-tuple whose diameter is checked to be 246.

6.2 Discovery-to-theorem chain

The source implements the following typed chain:

```
untrusted labelled JSON and search candidate
-> untrusted canonical generator
-> packed sparse tables
-> small Nat/Bool/RArray kernels
-> soundness bridges to direct mathematical recurrences
-> exact cleared-denominator inequality
-> explicit rational epsilon certificate
-> analytic L2 certificate
-> M_(50,1/25) > 4
-> Bombieri-Vinogradov and admissible H50
-> infinitely frequent prime gaps <= 246
```

The most transferable feature is not the particular recurrence. It is the refusal to trust the candidate generator, combined with local soundness lemmas that recover the mathematical specification from compact stored data.

6.3 Selected theorem and lemma map

Lean declaration	Role in the chain	What remains external or contextual
EpsCertificateExplicit	Finite rational coefficient family with a strict quadratic-form inequality.	The structure is specific to the sieve variational problem.
certFactorTables_correc	Connects cached factor tables to their declared mathematical factors.	It says nothing about PID event masses or lattices.
certStoredInequality	Kernel-decided strict inequality over stored row totals.	Stored-row correspondence needs separate lemmas.
certLhsStoredRowSum_eq, certRhsStoredRowSum_eq	Recover the original left and right quadratic forms from stored rows.	They share the certificate interpretation and source data.
certInequality	Produces the complete cleared-denominator witnessing inequality.	It is still a finite certificate predicate.
EpsCertificateExplicitPrecomputedInt.toExplicit	Bridges packed exact data and direct moments to the explicit rational certificate.	Integer runtime and soundness premises remain in the trust analysis.
EpsCertificateExplicit.toAbstract	Converts the explicit rational polynomial into the analytic L2 certificate.	This bridge is prime-gap-specific; only its architectural role transfers.
EpsCertificate.four_lt_MEps	Derives the strict variational inequality from the analytic witness.	It does not provide the Bombieri--Vinogradov premise.
existsEpsCert50	Packages the checked certificate as the endgame hypothesis.	The existential is only one premise of the final theorem.
lem_theta_bv	Uses Bombieri--Vinogradov to choose a certificate-admissible distribution level.	This is the analytic black box of the 246 development.
card_H50, diameter_H50, admissible_H50	Kernel-check the explicit 50-tuple, its diameter, and admissibility.	They do not prove the comment's "narrowest" optimality claim; the tuple has no PID analogue.
thm_main	Combines the analytic and finite routes to obtain frequent gaps at most 246.	The conclusion remains conditional on BombieriVino gradov.

Exact source anchors are retained in the transfer ledger. The theorem table is an architectural map, not an independent proof audit of every imported lemma.

7 The proof document and interactive graph

The blueprint is a readable 132-page mathematical document. Visual inspection of every page at contact-sheet scale and representative pages at higher resolution found consistent margins, legible formulas, stable headers, and useful per-statement Lean annotations. This is a substantial review aid.

The interactive paper contains 255 typed entities: 126 marked formalized, 21 marked gap, and 108 marked unknown. The graph records 1,126 typed uses: 805 statement uses and 321 proof uses. The observed dependency edges are acyclic and backward-pointing, with a maximum depth of 27 to the final theorem. This makes the site useful for locating high-centrality definitions and incomplete cuts.

The graph is not, in its present form, a durable correspondence certificate:

- 186 observed repository source links use mutable blob/main URLs;
- status and link presence are not equivalent: some formalized or gap nodes lack links, while some unknown nodes have links;
- later paper sections are largely marked unknown despite substantial Lean source elsewhere in the repository;
- PaperTag.lean constructs its blueprint URL from a literal todo string; and
- its duplicate-handling path logs “skipping duplicate” but still inserts the duplicate entry; and
- no same-revision receipt binds the HTML, PDF, Lean tree, graph edges, statuses, and source spans.

The concept transfers: maintain a revision-pinned claim/proof/evidence DAG. The current mutable site implementation does not transfer as an assurance mechanism.

8 Repository and CI audit

PrimeGapsLib separates PrimeGapsTheory, PrimeGapsCert, and their composite library. That split is valuable because a reviewer can build or inspect the mathematical theory without always replaying the large computation.

The reviewed repository nonetheless has important reproducibility and custody limits:

- the reviewed commit's reachable history contains seven commits and begins with a large monolithic import; no release tag or GitHub release asset is bound into this report;
- PrimeNumberTheoremAnd is requested from mutable main, although the manifest records the resolved commit used by this checkout;

- the certificate JSON is transitively bound by the reviewed Git tree, but has no explicit sidecar content digest or candidate-generation/search receipt beyond that repository-object binding;
- the workflow uses mutable action version tags rather than full commit pins;
- the latest reviewed run restored roughly 2.47 GB of cache, built successfully, but skipped the comparator, leanchecker, and nanoda routes and uploaded no proof receipt;
- the comparator configurations are documented but not wired into the reviewed CI workflow; and
- the reviewed project metadata describes self-assessed and internal review, and this report binds no independent human or institutional review receipt.

None of these observations refutes the Lean theorem. They limit claims about fresh reproducibility, external verification, review independence, and durable publication.

One parser detail illustrates why even a good architecture cannot be copied mechanically. The PrimeGaps source generator converts some labelled JSON integers using `Int.toNat`, so a negative input can be coerced rather than rejected. For an existential witness theorem this can be safe: the downstream checker only needs the reconstructed object to be *some* valid witness. A PID report, by contrast, is a deterministic function of the exact experimental count table. Its decoder must reject such coercions, require canonical bytes, and bind the accepted data identity.

9 Twenty-lens council

The council is a structured attack, not a vote. Each lens asks a different question of the same proposed transfer. Agreement among lenses does not create evidence, and a single load-bearing objection can veto promotion. The detailed evidence and nonimplications are preserved in the machine-readable transfer ledger. The table below uses the same canonical names, numbering, and order as the repository-wide 20-lens registry; it is the human synthesis of those exact rows.

#	Lens	PrimeGaps observation	PID decision
1	Estimand	The public result concerns bounded gaps between primes.	Name the deterministic averaged empirical categorical SxPID3 functional; transfer proof roles only, never sieve objects or constants.
2	Types	One fixed existential rational polynomial witness suffices for the source theorem.	PID accepts bounded runtime count tables, so canonical bytes, row roles, semantic masses, exact plans, intervals, and dispositions require distinct types.
3	Quantifiers	The public theorem retains an explicit Bombieri-Vinogradov premise and an existential certificate.	State the finite-input SxPID3 implication and its quantifier order; do not replace it with population, estimator, asymptotic, or causal prose.
4	Mappings	No theorem maps GPY/Maynard objects to shared-exclusions PID.	Transfer assurance roles only; any mathematical value or proposition needs a positive premise-checked target map.
5	Support/reference measure	PrimeGaps' analytic measure premises are internal to its simplex and sieve problem.	Reconstruct positive empirical support from the PID definition; do not infer continuous support, a reference measure, or KSG validity.
6	Lattice/source count	Packed signatures and recurrence indices form source-specific carriers.	Freeze the 18-node three-source antichain poset, its order, event semantics, and Möbius direction independently.
7	Units/gauge	The certificate compares dimensionless exact rational forms.	Preserve natural logarithms, nats, informative/misinformative/net orientation, normalization, and any later preprocessing gauge.
8	Population/empirical	The source certificate is an exact finite witness inside a number-theory proof, not a statistical estimate.	Keep empirical PMF functional, estimator output, fixture, represented binary value, and population quantity separate.
9	Sampling	The fixed prime-gap witness does not supply PID row-law assumptions.	The narrow exact categorical claim is deterministic; any later population, bootstrap, permutation, or confidence statement needs its own sampling contract.
10	Selection/UQ	Numerical search found a successful candidate, but the reviewed project lacks a complete exposure, stopping, and rejected-candidate ledger.	Freeze objectives and judges before candidate access, retain adaptation history, and never treat rediscovery or consensus as calibration.
11	Formal correspondence	Lean bridges explicit objects, but the website graph has mutable links and status inconsistencies.	Bind a reviewed paper-to-formula map for event, averaging, order, units, and conclusion before formal code earns semantic credit.
12	Kernel/axioms/toolchain	The project pins Lean and resolved dependencies, while reviewed CI skips comparator, leanchecker, and nanoda.	Inventory kernel/runtime/import closure/axioms and require fresh fail-closed replay; sorry, theorem premises, conventional axioms, and noncomputable are different categories.

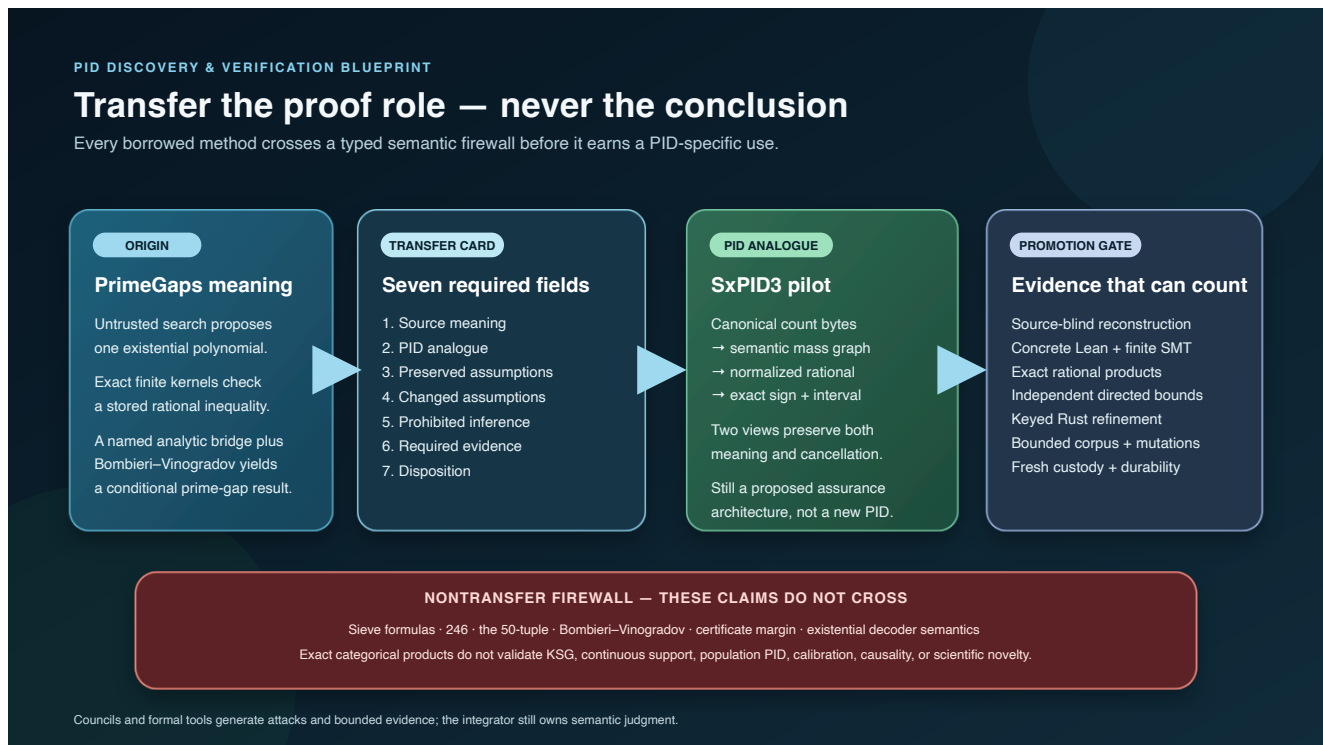
#	Lens	PrimeGaps observation	PID decision
13	Route dependency diversity	Theory, certificate, packed kernels, and optional comparator diversify some cuts but share source mathematics and data.	Record shared semantic, parser, arithmetic, custody, and model cuts; route or agent count is not independence.
14	Numerical/binary64	Approximate search is replaced by an exact rational acceptance inequality.	Use exact products for finite sign and separately verified directed intervals for log magnitude; neither validates binary64 estimators or platform libm.
15	Compiled/wrapper parity	The reviewed certificate route does not exercise pid-rs Rust/Python surfaces.	Compare all 108 stable keys across specialized/general, serial/parallel, feature, build, platform, serialization, and Python routes.
16	Counterexample/mutation	Small exact kernels are valuable only if load-bearing premises actually affect acceptance.	Mutate decoding, event connectives, target restriction, count weighting, lattice order, products, intervals, resources, claims, and coordinated reseals.
17	Custody/threat/refusal	Public source and a warm-cache build do not provide an external-custody comparator or certificate receipt.	Freeze candidate scope and judge authority, refuse unavailable verification, and retain exact acquisition/execution records plus same-actor limits.
18	Citations/novelty	PrimeGaps cites its mathematical sources and presents a formalization.	Pin primary statements and application hypotheses; classify the PID assurance design as project-defined engineering, not a new measure or theorem.
19	Ecosystem/authority	The interactive graph is useful navigation metadata, not an acceptance authority or proof-term DAG.	Bind claim, source, theorem, executable, consumer, and archive authority directions at one revision.
20	Resource/platform/release	Packed computation is fast in a warm cached run, but no reviewed same-revision release receipt closes the lane.	Start with a bounded streaming route; freeze resource refusal, attempts, withdrawals, OS/toolchain scope, release assets, durability, and recovery before promotion.

9.1 Council synthesis and dissent

- **Strongest surviving mechanism:** untrusted candidate generation followed by exact reconstruction in a smaller checker and a named semantic bridge.
- **Strongest objection:** identical words such as *factor*, *certificate*, *positive*, or *lattice* can conceal different objects and different sufficient conditions.
- **Most likely false assumption:** that two nominally independent routes do not share the same paper transcription, generated registry, parser, arithmetic library, model lineage, or custody.
- **Evidence that would change the decision:** a pinned primary-source map; a source-blind direct exact implementation; concrete Lean and separately encoded finite-SMT routes; complete hostile models; dependency-disjoint interval reconstruction; keyed Rust refinement; and a fresh, externally held replay.
- **Dissent retained:** the semantic DAG may be unnecessary. A simpler direct reconstruction is the baseline and remains preferred unless profiling and proof size show a measured benefit.

10 The transfer ledger: adopt structure, not conclusions

Every row in the companion ledger has source semantics, a PID analogue, preserved assumptions, changed assumptions, prohibited inferences, required evidence, and a disposition. The decisions compress to the following map.



10.1 Accepted architectural transfers

1. **Theory / certificate / composite separation.** The theory lane states general categorical semantics. The certificate lane checks one finite input. The composite lane is the only lane allowed to state the bound end-to-end implication.
2. **Untrusted discovery / small checker.** Search and generation remain replaceable. Acceptance reconstructs the object from canonical bytes rather than trusting a producer's tables.
3. **Finite-to-semantic bridges.** Stored products, matrices, or graph nodes earn meaning only through checked evaluation and refinement statements.
4. **Premises in types and reports.** Theorem inputs and report fields preserve units, support, estimand, source, and conditional status.
5. **Content-addressed promotion.** Exact inputs, outputs, tools, and evidence move together under one same-revision manifest.

10.2 Conditional transfers

- A human challenge and comparator are useful only when executed with pinned binaries, theorem and import rosters, axiom policy, raw results, and hostile tests.
- A proof-document DAG is useful only when revision-pinned and checked for unique identities, valid spans, status coherence, and evidence direction.
- Sparse graphs, packing, and sharding are implementation choices. They require completeness proofs and benchmarks; they are not mathematical evidence by themselves.
- Autoresearch may adaptively search, but a candidate-inaccessible judge and a complete exposure ledger control promotion.

10.3 Rejected transfers

- No GPY or Maynard expression, eigenvector objective, admissible tuple, or number 246 transfers to PID without a new theorem establishing an actual typed map.
- No exact categorical product validates KSG, continuous shared exclusions, population PID, or a statistical procedure.
- No warm-cache build, self-review, agent consensus, mutable website, or polished PDF supplies independent acceptance.

11 PID target: averaged empirical categorical SxPID3

The chosen pilot is narrow because it has exact finite semantics and a real open assurance packet. It is **not** the full mixed-dimensional continuous PID3, not KSG, not fitted quantization, and not pointwise certification. The existing packet remains proposed and all five closure programs are open.

Let a supported joint realization be

$$z = (s_1, s_2, s_3, t), \quad c_z \in \mathbb{N}_{>0}, \quad N = \sum_z c_z > 0.$$

An antichain α contains nonempty source-index sets. For the keyed row z , define the source event

$$E_\alpha(z) = \bigcup_{a \in \alpha} \bigcap_{i \in a} \{S_i = s_i\}.$$

The union across antichain branches and conjunction within a branch are load-bearing. Write the target event as $T(z) = \{T = t\}$. Define

$$\begin{aligned} U_{\alpha,z} &= \sum_{z' \in E_\alpha(z)} c_{z'}, \\ V_{\alpha,z} &= \sum_{z' \in E_\alpha(z) \cap T(z)} c_{z'}, \\ T_z &= \sum_{z' \in T(z)} c_{z'}. \end{aligned}$$

For valid supported rows these masses are positive. The informative, misinformative, and net local ratios are

$$\rho_{\alpha,z}^+ = \frac{N}{U_{\alpha,z}}, \quad \rho_{\alpha,z}^- = \frac{T_z}{V_{\alpha,z}}, \quad \rho_{\alpha,z}^{\text{sx}} = \frac{\rho_{\alpha,z}^+}{\rho_{\alpha,z}^-} = \frac{NV_{\alpha,z}}{U_{\alpha,z}T_z}.$$

Empirical weighting is by c_z/N , not uniformly over distinct supported rows. Clearing the logarithmic sums into exact positive-rational products gives

$$\begin{aligned} Q_\alpha^+ &= \prod_z \left(\frac{N}{U_{\alpha,z}} \right)^{c_z}, \\ Q_\alpha^- &= \prod_z \left(\frac{T_z}{V_{\alpha,z}} \right)^{c_z}, \\ Q_\alpha^{\text{sx}} &= Q_\alpha^+ / Q_\alpha^-, \\ C_\alpha^u &= \frac{1}{N} \ln Q_\alpha^u, \quad u \in \{+, -, \text{sx}\}. \end{aligned}$$

There are exactly 18 nonempty antichains of the seven nonempty subsets of three sources in the frozen registry. The redundancy order is

$$\alpha \preceq \beta \iff \forall b \in \beta \exists a \in \alpha : a \subseteq b.$$

With cumulative rows and atom columns,

$$Z_{ij} = \mathbf{1}[\alpha_j \preceq \alpha_i], \quad C_i^u = \sum_j Z_{ij} \Pi_j^u.$$

The frozen Z has 129 ones. Its exact inverse $M = Z^{-1}$ has 65 nonzero entries in $\{-1, 0, 1\}$. The frozen acceptance contract checks both $MZ = I$ and $ZM = I$ as defense-in-depth; over the exact square rational matrices, either one-sided inverse identity already implies the other. Neither direction establishes event semantics: a consistently transposed or wrongly ordered pair can remain algebraically invertible.

Atom products and values are

$$R_i^u = \prod_{j=0}^{17} (Q_{\alpha_j}^u)^{M_{ij}}, \quad \Pi_i^u = \frac{1}{N} \ln R_i^u.$$

Because every R_i^u is a positive rational and $N > 0$, exact integer cross-multiplication decides

$$\Pi_i^u = 0 \iff R_i^u = 1, \quad \Pi_i^u > 0 \iff R_i^u > 1, \quad \Pi_i^u < 0 \iff R_i^u < 1.$$

A nonempty product may equal one. Exact zero is therefore not syntactic emptiness and not a floating tolerance. Magnitudes require a separate outward interval for $(1/N) \ln R$; interval overlap is not enough, and an interval touching zero does not decide exact sign.

The registry contains 108 coordinates: 18 cumulatives and 18 atoms for each of informative, misinformative, and net. The bounded binary qualification corpus has

$$\sum_{n=1}^5 \binom{n+15}{15} = \binom{21}{16} - 1 = 20,348$$

labeled count vectors and $20,348 \times 108 = 2,197,584$ coordinate verdicts. These are not all arbitrary-alphabet inputs and not 20,348 distinct probability laws; only 20,164 are primitive rational laws after division by the count-vector gcd.

12 Proposed proof-carrying semantic IR

The strongest prospective transfer is a two-view intermediate representation. It generalizes the existing certified-sxpid lowering pipeline to keyed SxPID3; it is not a new PID measure, and it is not yet a demonstrated engineering improvement.

12.1 View 1: semantic provenance graph

Each node retains a role-labelled key rather than only a number:

```
Total(N)
TargetMass(target=t)
EventMass(alpha, supported_row)
TargetEventMass(alpha, supported_row)
LocalRatio(component, alpha, supported_row)
Cumulative(component, alpha)
Atom(component, antichain_key)
```

The row count c_z is an exponent and provenance weight, not another multiplicative base. The graph records why a mass exists, which event produced it, which source/target values key it, and which coordinate consumes it. Numeric equality does not merge different semantic roles.

Required graph obligations are bidirectional:

1. **Forward completeness:** every mass and factor required by the frozen formulas appears exactly once under its semantic key.
2. **Backward soundness:** every stored node corresponds to one valid required object; there are no unreferenced or invented factors.
3. **Evaluation commutation:** evaluating the graph from the decoded count table yields the direct event/product specification for every key.
4. **Permutation equivariance:** renaming sources transports keys and values through one of the six source permutations without positional assumptions.

12.2 View 2: normalized exact arithmetic

Semantic keys are not a canonical arithmetic normal form. For example,

$$\frac{2 \cdot 6}{3 \cdot 4} = 1$$

despite a nonzero raw exponent vector over bases 2, 3, 4, 6. Therefore the acceptance route evaluates the semantic graph into a normalized positive rational, or compares its exact numerator/denominator cross-products after resource preflight. Structural cancellation may be a fast sufficient proof of zero; it is never the exact-zero authority.

Conversely, merging nodes only because their values happen to be equal would erase semantic mutation visibility. A target mass of six and an event mass of six are equal numbers but different claims. The two views solve different problems:

View	Preserves	Must not be used as
Semantic graph	roles, keys, provenance, completeness, mutation observability	canonical rational equality or exact sign authority
Normalized rational/cross-product	arithmetic cancellation, equality with one, strict sign	proof that factors came from the correct events or data

12.3 Baseline and adoption gate

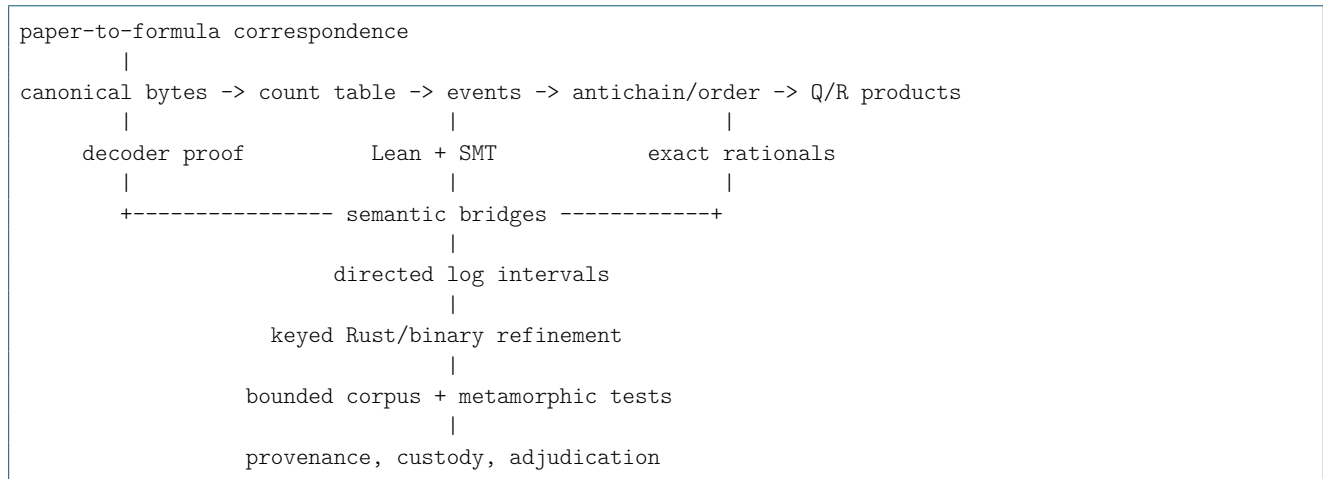
audit/tools/certified-sxpid/src/exact.rs, extract.rs, and product.rs already provide the broad two-source pattern: canonical exact log arguments, count/event lowering, Möbius application, resource preflight, exact rational powers, and comparison with one. The first SxPID3 implementation must extend or independently reconstruct that simple direct route. The DAG is adopted only if it:

- closes a named Program C or D obligation without weakening semantics;
- is proved equivalent to direct reconstruction;
- reduces measured time, peak memory, certificate size, or review complexity on the exact corpus;
- keeps resource admission and public certificate semantics stable, or triggers a new claim revision when it does not; and
- survives the noncanonical-factor and semantic-role mutation suite.

Packed arrays and custom bit-level kernels remain deferred until profiling shows that this clearer route is inadequate.

13 Layered assurance architecture

The architecture separates claims that are often collapsed into the phrase “formally verified.”



No downstream green box repairs an upstream semantic error. The five closure programs are:

Program	Question	Main tools	Explicit nonclaim
A: semantic/combinatorial	Did the paper-defined event and lattice get transcribed correctly?	pinned source map, source-blind derivation, exact minimal witnesses	examples and citations are not a universal bridge
B: dual formal	Are carrier, order, matrix, event, product, and sign theorems derivable?	Lean concrete route; independently encoded SMT finite route	two kernels can share the same wrong source transcription
C: certified numerics	Do exact products and independent outward log intervals agree?	exact integers/rationals, MPFR/Arb producer, rational-series verifier	deterministic intervals are not confidence intervals
D: compiled refinement	Do the shipped Rust paths implement the keyed specification?	stable-key traces, serial/parallel/features/platform matrix	float agreement is not exact-product proof

E: adversarial replay	Are bytes, dependencies, mutations, first results, and custody complete?	content manifests, fresh replay, hostile fixtures, independent adjudication	hashes and reruns are not scientific truth
--------------------------	--	--	--

All five remain open for SX-CERTIFIED-AVERAGED-PID3-001. This report changes the design proposal; it does not close the claim.

14 Formal methods as bounded instruments

Formal verification is most valuable when it narrows a specific uncertainty and least valuable when it becomes a ceremonial shield. Every evidence record should name its trusted computing base, independence dimensions, finite bounds, residual risks, prohibited inference, and the weakest plausible false claim it could still accept.

14.1 Lean

Lean can establish that a term inhabits the encoded proposition under its imports and axiom dependencies. It cannot by itself establish that the proposition is the intended paper theorem, that runtime bytes denote the intended data, or that an estimator is scientifically realistic.

Four notions must remain separate:

1. hBV : BombieriVinogradov is a theorem premise;
2. propext, Classical.choice, and Quot.sound are conventional proof-environment dependencies reported by an axiom audit;
3. intentional sorry placeholders in challenge modules are holes in a human comparator target, not holes in the default production theorem libraries; and
4. noncomputable marks an execution/extraction property, not an unproved or unsound theorem.

14.2 SMT

For finite masks, antichains, orders, and integer matrices, SMT is a strong second encoding. A theorem route proves unsatisfiability of the negation; each hostile mutation should become satisfiable with a retained model. unknown is no evidence. A wrong but self-consistent encoding can still yield unsat, so the SMT route starts from raw Boolean/set constraints rather than a generated Lean registry.

14.3 Exact arithmetic and directed intervals

Exact rational products eliminate rounding only for the encoded finite expression. They still rely on correct decoding, events, ordering, resource accounting, and big-integer execution. Directed log intervals rely on correct range reduction, tail bounds, endpoint rounding, and runtime behavior; they may legitimately abstain. They are deterministic enclosures, not statistical confidence intervals.

14.4 Exhaustive, mutation, metamorphic, and differential testing

- Exhaustive enumeration proves behavior only over its declared finite carrier.
- Mutation testing demonstrates sensitivity only to named faults.
- Metamorphic relations can be shared by multiple wrong implementations.
- Differential routes are useful only after common code, libraries, generated data, and semantic premises are mapped.
- A green CI run is one environment/revision observation. A warm-cache build is not a cold reproducibility receipt.

14.5 Hashes, councils, and paper graphs

A digest binds bytes up to collision risk; it does not prove truth, authorship, authenticity, meaning, or durable availability. A council generates attacks and alternative designs; shared models, prompts, sources, tools, or custody make it correlated. A paper graph or status badge is curation metadata until a same-revision checker binds entities, declarations, spans, imports, and evidence.

15 Attack matrix and required negative controls

The acceptance predicates must fail for the intended reason under at least the following classes.

Cut	Mutation or counterexample	Required observation
Decoder	duplicate member, noncanonical integer, negative-to-natural coercion, Unicode alias, reordered canonical bytes	rejection before event work
Event	OR/AND swap, OR within a mask, missing target intersection	exact retained count witness disagrees
Weighting	uniform supported-row weight instead of exponent c_z	count-weighting witness rejects
Lattice	missing antichain, padded zero mask, reversed order, transposed Z , altered Möbius row	model or exact inverse/semantic reconstruction fails
Keying	positional (3)/(4) swap, source permutation without key transport	stable-key comparison rejects
Component	informative/misinformative swap, net multiplication instead of division, negative net clamp	exact quotient and reconstruction reject
Units	omit $1/N$, label bits as nats	identity and interval relation reject
Factor IR	$(2 \cdot 6)/(3 \cdot 4) = 1$ with nonzero exponent vector	normalized rational says zero; structural-only route fails
Factor roles	equal-valued target/event masses merged, or c_z treated as a base	semantic graph completeness/soundness fails
Product	nonempty product one, factor/exponent mutation, huge projected power	exact comparison or preflight rejects
Interval	overlap substituted for containment, endpoint sign, mutated $\ln 2$, same-backend “independent” route	verifier rejects or abstains
Registry	partial 54-coordinate report, reordered identity, self-consistently resealed wrong registry	complete 108-key binding rejects
Corpus	replicated tables called distinct laws; missing or duplicate shard range	ordinal/coverage checker rejects
Refinement	specialized/general paths compared positionally or a feature path omitted	keyed execution matrix rejects
Interpretation	exact empirical sign called significance, calibration, or population truth	wording/schema gate rejects

16 Autoresearch without evidence laundering

Autoresearch is a candidate generator. It can be ambitious, adaptive, and computationally broad, but promotion rules depend on the question being asked.

16.1 Four distinct research loops

1. **Theorem discovery.** Models or search produce lemmas, proof outlines, witnesses, or counterexamples. Promotion requires a kernel-checked theorem plus source-correspondence review.
2. **Exact finite PID falsification.** Search produces canonical integer count tables that maximize disagreement, cancellation, resource use, or mutation sensitivity. Promotion requires independent exact reconstruction of the table and predicate.
3. **Estimator design.** Search proposes algorithms or hyperparameters. Promotion requires preregistered data-generating families, held-out and repeated confirmation, uncertainty, multiplicity control, robustness, and selection-aware reporting. An exact finite witness cannot substitute for these.
4. **Formal proof generation.** A model proposes terms or tactics. Kernel acceptance checks the formal proposition; human and source-mapping review still own intended meaning.

16.2 Promotion ledger

Before a run, record:

- objective and prohibited proxies;
- candidate grammar and resource budget;
- seed, environment, code revision, and allowed tools;
- stopping and retention rules;
- judge identity and the candidate's access to it; and
- confirmation data or theorem surface held back from adaptive selection.

During search, retain every exposure that can influence later choices, not only winners. After search, freeze the candidate bytes before judging. The judge reconstructs from source input, emits one terminal result, and cannot be rewritten or queried adaptively by the candidate lane.

The research record keeps failures, near misses, exploit attempts, judge disagreements, and the first accepted result. A later clean rerun does not erase the selection history. Search frequency, score, model consensus, and repeated rediscovery are discovery signals, not evidence of the claim.

16.3 Council protocol

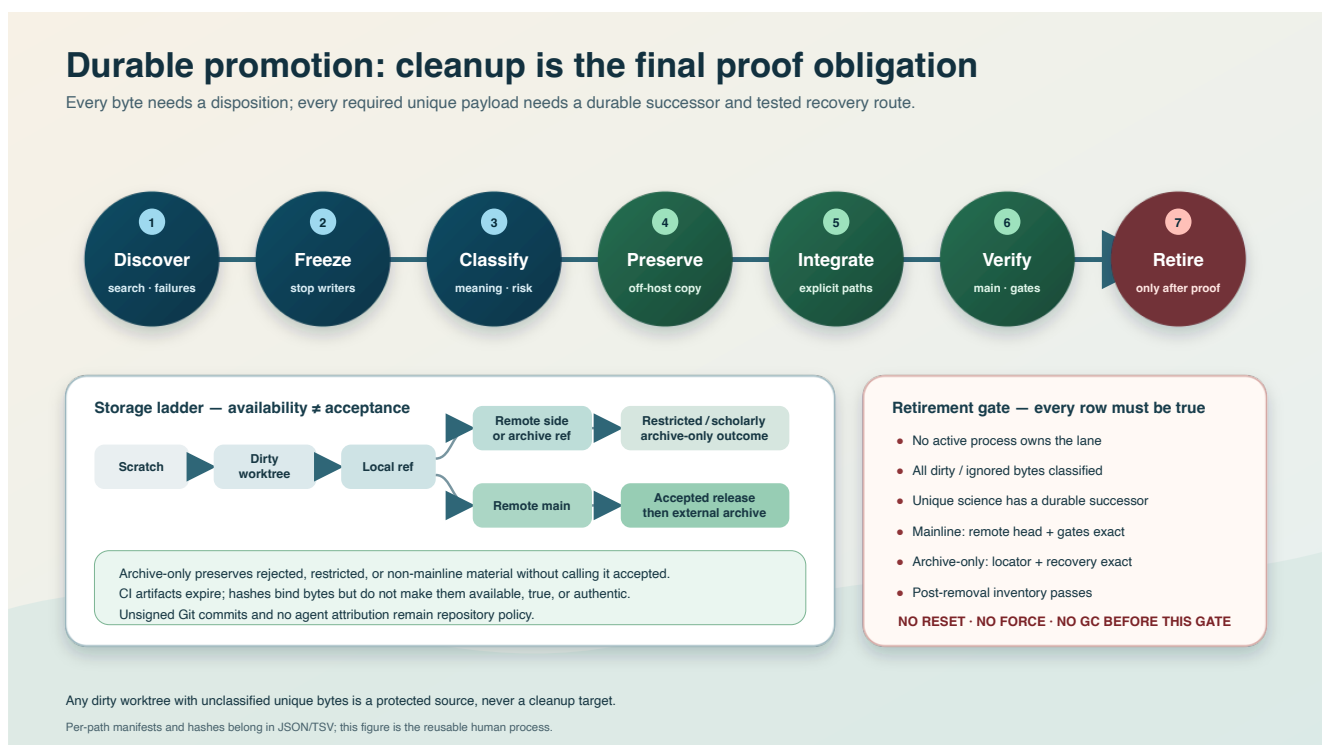
For consequential designs, twenty lenses are applied in three passes:

1. independent pre-merge memos identify the strongest falsifier and most likely hidden shared cut;
2. an integrator reconstructs decisive facts from primary sources and synthesizes competing designs; and
3. an adversarial pass attacks the selected design, its evidence boundaries, and its wording.

The record preserves dissent, the strongest surviving objection, evidence that would reverse the decision, and independence limitations. There is no majority vote and no “assurance percentage.”

17 Repository durability and promotion

Simple migrations should follow one reusable state machine rather than produce a bespoke PDF. The human overview lives here; exact per-path facts belong in a JSON/TSV cleanup ledger.



Discover → Freeze → Classify → Preserve → Integrate → Verify → Retire

17.1 Storage and trust ladder

Layer	Useful for	Not sufficient for
Scratch file / model context	rapid exploration	recovery, review, or durable identity
Dirty worktree	active local development	evidence that all bytes are tracked or backed up
Local commit/ref	recoverable local history	off-machine durability
Remote side/archive ref	off-host exact recovery	accepted mainline status or scholarly permanence
Remote main	integrated project history	release identity or long-term external archive
Signed-off project release metadata, unsigned Git objects per repository policy	named immutable project milestone	independent scholarly deposit or authenticity by itself
DOI/scholarly or restricted archive	long-term citation/recovery under its policy	automatic scientific correctness
CI artifact	short-lived diagnostic exchange	archival storage; expiration is expected

The repository policy requires unsigned commits and no AI attribution. “Signed-off” in the table is therefore a

human decision concept, not a cryptographic Git signature.

17.2 Discover and freeze

Inventory every common Git directory, worktree, branch, tag, stash, checkpoint, unreachable object, dirty tracked path, untracked path, and relevant ignored path. Record HEAD, tree, merge base, mode, size, digest, and process owner. Stop writers before taking a final manifest. Do not assume `git status` covers ignored research archives, caches that contain unique results, or alternate object stores.

17.3 Classify and preserve

Classify each byte variant as accepted science, open candidate, retained negative evidence, superseded-but-needed recovery state, restricted/local-only material, reproducible cache, or discardable temporary output. Secret-screen before copying. Environment files, credentials, device IDs, tokens, and private paths never enter a public snapshot.

Unique or uncertain material receives an immutable local ref and an off-host, content-addressed snapshot or reviewed archive path before integration. A branch is not deleted because its name looks old; a worktree is not removed because its files resemble main.

17.4 Integrate

Start from fresh remote main, not from a dirty primary worktree. Port explicit paths or coherent claim packets. Reconcile changes by semantic role, not wholesale directory copy. Regenerate self-referential source states, receipts, and publication derivatives last. Keep commits small, unsigned, human-authored, and free of agent trailers.

For unpublished PID material, use separate reviewed successors for distinct scientific claims. Negative and withdrawn results remain reachable under explicit archive/evidence paths rather than being silently overwritten by accepted variants.

17.5 Verify and retire

Before push, require the relevant mathematical, formal, numerical, source-custody, build, test, lint, documentation, PDF, and mutation gates. After push, bind the exact remote head and hosted run identities. Local and remote main must resolve to the same intended commit without force.

A branch or worktree is retirement-eligible only when:

1. no process owns it;
2. every dirty, untracked, and relevant ignored byte has an explicit disposition;
3. its commits are remote-main ancestors, or every unique payload has a durable successor and recovery ledger;
4. restricted material has a safe non-public home;
5. the remote proof and recovery drill pass; and
6. a post-removal inventory shows no newly unreachable required object.

Cache deletion and Git garbage collection come last, after a retention window and repeated object inventory. Remote archive branches remain until exact preimages have a main-reachable or verified external successor.

17.6 Current safety disposition

At the review boundary, the dirty primary worktree contained substantial unpublished PID claim packets, scripts, reviews, and ignored material not proved redundant with remote main. It was therefore treated as a protected source, not a cleanup target. The SxPID3 packet used here was ported into this isolated lane with an exact preimage manifest, then received bounded presentation-only GitHub-Markdown normalization: legacy inline/display delimiters became $/$$, four `\operatorname` occurrences across three formulas became `\mathrm`, and one blockquoted implication was reflowed as ordinary prose plus display math. The ledger binds both preimage and current bytes. These mechanical edits were not intended to change mathematical or prose content, but the hashes alone do not prove semantic equivalence, and the preservation step does not adjudicate the packet. No cleanup or repository-wide integration claim is made here.$

18 Implementation roadmap for SxPID3

The order below optimizes for semantic risk reduction, not visual progress.

18.1 Phase 0: preserve and adjudicate the packet

- Manifest and secret-screen the complete SX-CERTIFIED-AVERAGED-PID3-001 packet.
- Port it by explicit path onto a reviewed main descendant without changing its proposed status.
- Pin the exact Makkeh-Gutknecht-Wibral source and construct a paper-equation-to-local-formula map.

- Decide whether the semantic DAG is internal revision-1 machinery or a claim-v2 acceptance object.

Stop condition: any unresolved source/event/unit/order ambiguity.

18.2 Phase 1: simple direct specification

- Implement the strict canonical decoder and direct event-count route.
- Reconstruct all 18 antichains, order entries, Z , M , 54 cumulative products, 54 atom products, and 108 stable keys without generated acceptance tables.
- Add minimal exact witnesses for every semantic mutation and the noncanonical-factor example.

Promotion condition: source-blind review and exact keyed results on hand-checkable fixtures.

18.3 Phase 2: dual formal semantics

- In Lean, define the concrete carrier, event membership, counts, order, zeta/Möbius relation, products, and sign reduction.
- Independently encode finite masks, antichains, order, all 324 zeta entries, inverse products, permutations, and event mutations for SMT.
- Freeze theorem/import/axiom rosters and run hostile substitutions.

Nonclaim: this does not close paper correspondence, log intervals, or Rust execution.

18.4 Phase 3: exact products and intervals

- Build a producer and dependency-disjoint verifier from canonical count bytes.
- Compare normalized exact rationals with one after aggregate/per-coordinate preflight.
- Produce directed intervals and require the verifier interval to be a subset of the producer interval.
- Make precision exhaustion and resource refusal terminal, with no partial verified result.

18.5 Phase 4: keyed Rust refinement

- Expose audit-only stable-key traces from specialized and general three-source routes.
- Compare serial/parallel, debug/release, features, platforms, package/binary identity, and Python exposure without positional assumptions.
- Keep binary64 tolerance results diagnostic; exact products and intervals remain the authority.

18.6 Phase 5: bounded corpus and shards

- Freeze a canonical ordinal for all 20,348 count vectors.
- Partition half-open ranges into content-addressed shards with no overlap or gap.
- Replay all 2,197,584 coordinate verdicts plus permutations, relabelings, replication, canonical gates, retained failures, cancellation, and resource refusals.
- Prove the shard root corresponds to the exact ordinal domain; do not call it arbitrary-alphabet exhaustiveness.

18.7 Phase 6: promotion and external review

- Freeze source, tools, schemas, binaries, first-result records, negative evidence, and mutation receipts under one same-revision manifest.
- Obtain fresh acquisition and execution under separate custody.
- Adjudicate every Program A--E obligation. Any load-bearing open cut keeps the claim proposed or blocked.
- Publish one milestone MD/PDF and machine companions; do not create migration-specific narrative PDFs.

19 Evidence-card template

Every acceptance-bearing artifact should answer:

Field	Required content
Claim	Exact proposition and quantifiers supported by this item
Source semantics	What the originating object means in its own field
Bound	Input domain, count, precision, resource, platform, and revision limits
TCB	Kernels, runtimes, parsers, libraries, solvers, filesystem, and hardware assumptions
Independence vector	Different acquisition, custody, language, encoding, algorithm, backend, and reviewer dimensions
Shared cuts	Paper transcription, registry, parser, generator, arithmetic, data, or tooling dependencies
Defeaters	Conditions under which valid-looking evidence no longer supports the claim
Weak false claim accepted	The most plausible wrong proposition this route could still pass

Field	Required content
Prohibited inference	What downstream statement must not cite this item
Reproduction	Exact inputs, command, outputs, hashes, and terminal status
Durability	Remote/release/archive locator, retention policy, and recovery test

20 Final recommendation

PrimeGapsLib offers a strong architectural lesson: a sophisticated search may remain outside the trusted argument if exact finite data are reconstructed by a smaller checker and connected to the final theorem through explicit bridges. It also offers a warning: elegant formal code, a large blueprint, a public build, and an interactive graph can coexist with mutable links, unexecuted comparators, warm-cache evidence, sparse process provenance, and no same-revision publication receipt.

For `pid-rs`, the best next step is not to imitate prime-gap mathematics or immediately build a packed kernel. It is to preserve the open `SxPID3` packet, close the paper-to-formula map, implement a simple source-blind exact route, and pilot a two-view proof-carrying semantic IR whose semantic and arithmetic obligations are separately testable. Only measured benefit justifies packing or sharding. Only Programs A--E justify a certificate disposition. Nothing in that path upgrades KSG, continuous PID, population inference, or scientific interpretation without their own evidence.

The process recommendation is equally important: keep adaptive discovery broad, make promotion narrow, retain failures and selection history, expose shared cuts, generate one canonical human report from the same mathematical source, and move every valuable byte through an auditable storage-and-promotion ladder before retiring any branch or worktree.

21 Source-anchored claim register

The links below are pinned to the reviewed Git commit. They are locators for review, not a claim that a line range alone contains every transitive proof dependency.

Review claim	Primary anchor
Final public declaration is conditional	PrimeGaps/Bounded246.lean, lines 21--31
Stored certificate proves the cleared threshold corresponding to > 4	PrimeGapsCert/Gap246/Witness.lean, lines 35--46
4.004306 appears in a separate positivity-window theorem	Sieve/PositivityWindow.lean, lines 68--81
Explicit tuple card, diameter, and admissibility are checked	H50.lean, lines 31--50
Candidate generator is explicitly untrusted	Emit/SourceGen.lean, lines 10--15
Generator uses integer-to-natural conversion in labelled input decoding	Emit/SourceGen.lean, lines 69--89
Small kernel layer is isolated from the untrusted producer	Kernel/Core.lean, lines 9--14
Stored rows and the terminal inequality are reconstructed and packaged	Witness.lean, lines 19--46
Precomputed integer tables are converted to the explicit rational certificate	Certificate/Fast.lean, lines 4068--4134
The explicit rational certificate is bridged to the analytic L2 witness	ExplicitToAbstract.lean, lines 687--731
Human challenge and solution are separate roots	Challenge/Basic.lean, lines 8--48, Solution/Basic.lean, lines 1--3
Paper-tag URL and duplicate handling need hardening	PaperTag.lean, line 100 and lines 158--168
Reviewed public CI is a minimal tagged-action build	.github/workflows/lean.yml, lines 1--17
External PNT dependency is requested from mutable main	lakefile.toml, lines 20--23

The repository-wide search found six sorry placeholders, all in the two intentional challenge roots rather than the default theorem libraries. That fact does not turn challenge holes into production proof holes, nor does it establish that the challenge statements perfectly represent the paper. The comparator still needs an executed, pinned receipt

before it earns verification credit.

22 Primary references

- Axiom Math, PrimeGapsLib at reviewed commit.
- Axiom Math, interactive bounded-prime-gaps paper.
- Axiom Math, formalization blueprint PDF.
- Makkeh, Gutknecht, and Wibral, *Introducing a differentiable measure of pointwise shared information*, Physical Review E 103, 032149 (2021), arXiv:2002.03356.
- pid-rs, MATHEMATICAL_PROBLEM_SOLVING_WORKFLOW.md, the existing repository-wide scientific and autoresearch process record.
- pid-rs, claims/SX-CERTIFIED-AVERAGED-PID3-001/, proposed revision-1 specification and open Programs A--E.